Patterns In Youth Marijuana Use

Understanding Drug Usage in Minors through Decision Trees and Ensemble Methods

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Introduction

Marijuana is the most commonly used illicit drug by US adolescents. While the usage rates have remained stable over time in youth as legislation around marijuana has changed, the potential for more negative health outcomes still persist.

This project seeks to investigate which demographic and environmental factors are most related to the usage of marijuana in US teens under the age of 18.

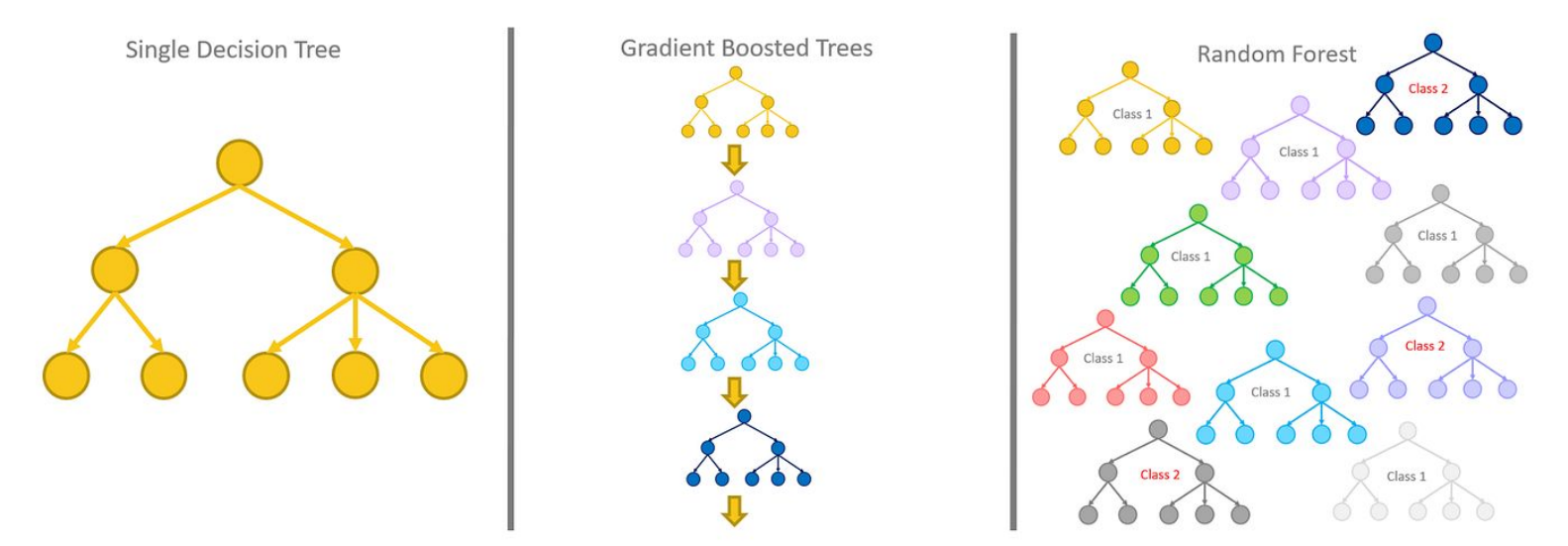
Introduction: Key Goals and Approach

- Can decision trees and ensemble methods (bootstrap aggregating, random forests, gradient boosting) be used to predict marijuana use in youth?
- Three approaches: **binary classification** (has used before vs. has not), **multi-class classification** (frequency of usage in the past year), **regression** (at what age did an individual begin using)
- Which approach and model generate the highest accuracy, and which features most strongly associate with youth cannabis use?

Introduction: Dataset

- National Survey on Drug Use and Health (2023) from Substance Abuse and Mental Health Services Administration
- +10,000 observations (respondents) across 79 features (survey questions)
- Features/predictors of interest: demography, parental influence, parental attitudes on drug usage

Theoretical Background



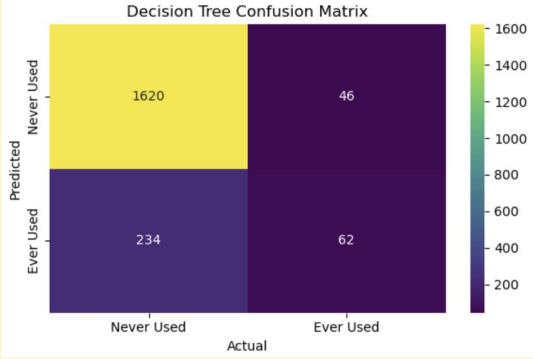
Silipo, R. (2020, March 19). *Ensemble models: Bagging & boosting*. Medium. https://medium.com/analytics-vidhya/ensemble-models-bagging-boosting-c33706db0b0b

Data Methodology

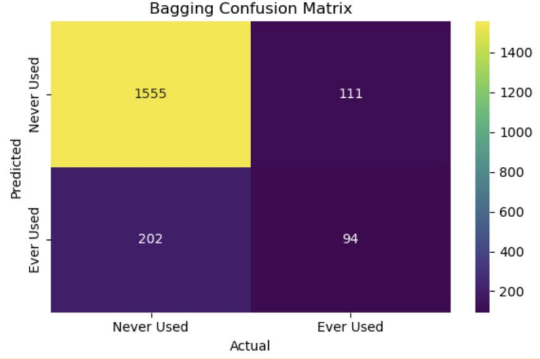
- Data filtered to include only responses for those under 18 (~30,000 observations to ~10,000)
- Variables narrowed down to 14 total predictors
- Three target variables of interest:
  - 'MRJFLAG': has or has not used marijuana
  - 'IRMJFY': frequency of use of marijuana in past year (generalized into three classes)
  - 'IRMJAGE': age of first use of marijuana (excluding all respondents who have never used marijuana)
- Data split using 80/20 train-test split, implemented all models and training data and tested for accuracy with new data
- Classification assessed using accuracy and confusion matrices; regression assessed using metrics such as MSE, RMSE, and R^2

Binary Classification Results

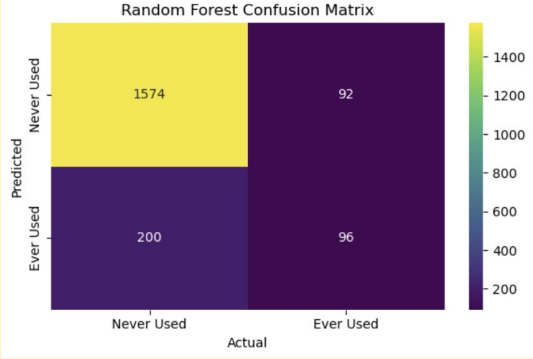
Decision Tree (accuracy: 85.73%)



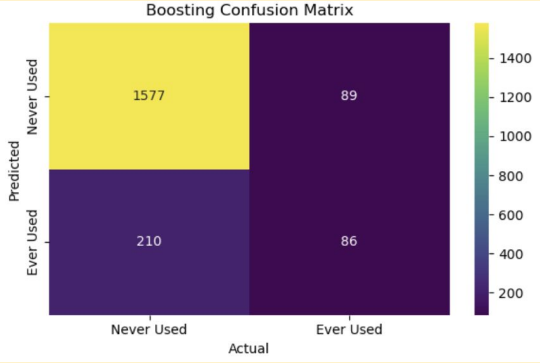
Bagging (accuracy: 84.05%)



Random Forest (accuracy: 85.12%)

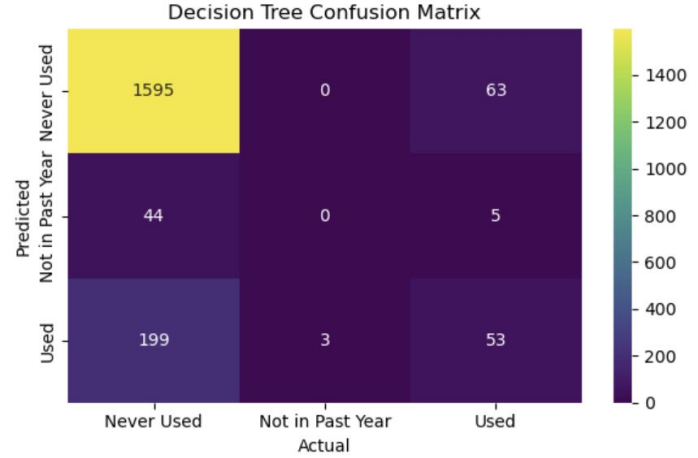


Boosting (accuracy: 84.76%)

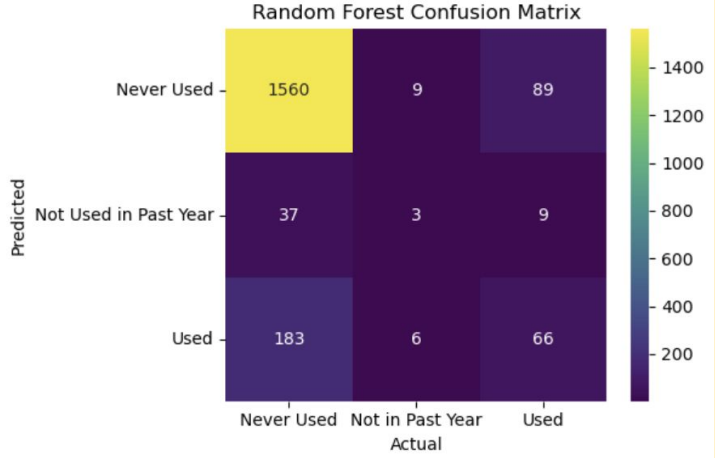


Multi-class Classification Results

Decision Tree (accuracy: 84.00%)



Random Forest (accuracy: 83.03%)



Regression Results

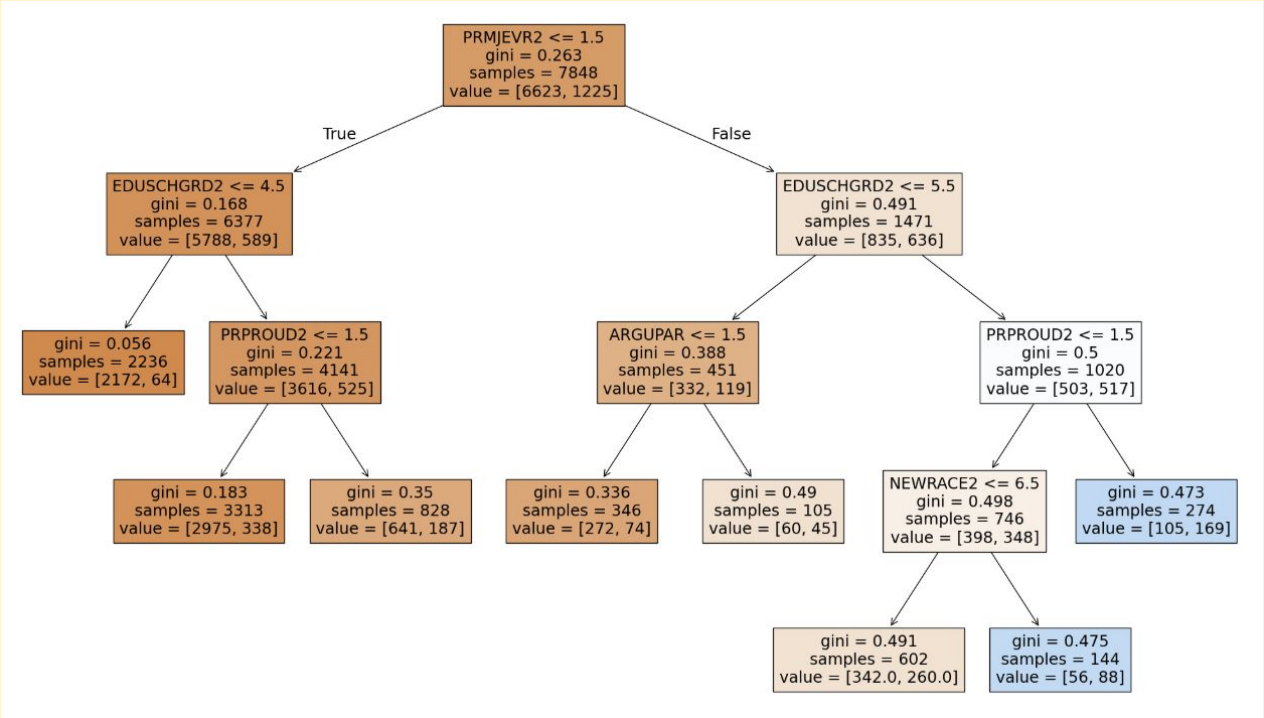
Model	MSE	RMSE	R^2
Decision Tree	2.6143	1.6169	0.1441
Boosting (λ = 0.1)	2.8267	1.6813	0.0746
Boosting (λ = 0.01)	2.4519	1.5659	0.1973

- Standard deviation of age of first use = 1.81
- Potential overfitting with higher shrinkage parameter

Discussion

- **Binary classification:** pruned decision tree achieved highest accuracy (85.73%) but detected the lowest proportion of those who had used marijuana correctly; random forest was second best (85.12% accuracy)
- **Multi-class classification:** pruned decision achieved highest accuracy (84.00%) but also struggled with minority class; decision tree accuracy: 83.03%
- **Regression:** boosting model with optimal shrinkage parameter achieved highest accuracy (MSE: 2.45)

Discussion: Pruned Decision Tree



Discussion: Feature Importance

	feature_name	importance
0	PRMJEVR2	0.716815
1	EDUSCHGRD2	0.168382
2	PRPROUD2	0.076177
3	NEWRACE2	0.019363
4	ARGUPAR	0.019264
5	IRSEX	0.000000
6	INCOME	0.000000
7	COUTYP4	0.000000
8	RLGATTD	0.000000
9	PRVDRGO2	0.000000
10	PARCHKHW	0.000000
11	PARHLPHW	0.000000
12	PRLMTTV2	0.000000
13	PARLMTSN	0.000000

Binary Classification

	feature_name	importance
0	PRMJEVR2	0.620882
1	EDUSCHGRD2	0.189464
2	PRPROUD2	0.063134
3	ARGUPAR	0.031067
4	NEWRACE2	0.024636
5	INCOME	0.018015
6	COUTYP4	0.016527
7	PRLMTTV2	0.015318
8	RLGATTD	0.009664
9	IRSEX	0.005648
10	PARCHKHW	0.005647
11	PRVDRGO2	0.000000
12	PARHLPHW	0.000000
13	PARLMTSN	0.000000

Multi-class Classification

	feature_name	importance
0	EDUSCHGRD2	0.446242
1	NEWRACE2	0.106175
2	COUTYP4	0.064711
3	INCOME	0.061793
4	PRMJEVR2	0.045504
5	ARGUPAR	0.043383
6	PRPROUD2	0.035183
7	RLGATTD	0.034717
8	PARHLPHW	0.033811
9	PARLMTSN	0.031634
10	PARCHKHW	0.030298
11	PRLMTTV2	0.029447
12	IRSEX	0.025538
13	PRVDRGO2	0.011564

Regression

Discussion: Potential Limitations

- Class imbalance in both formulations of classification
- Grade level is potentially redundant in the regression problem
- Youth responses to the survey may be inaccurate
- Broad amount of factors that could also contribute to youth marijuana use

Conclusion

- Parental attitudes towards drug use appear to have the strongest assocation with US youth usage
- Grade level, race, and income also demonstrate association
- Ensemble methods offer more thorough predictive capability, at the occasional cost of accuracy
- Future research built on these methods to deal with class imbalance

Bibliography

Hammond, C. J., Chaney, A., Hendrickson, B., & Sharma, P. (2020). Cannabis use among U.S. adolescents in the era of marijuana legalization: a review of changing use patterns, comorbidity, and health correlates. *International review of psychiatry (Abingdon, England)*, *32*(3), 221–234. https://doi.org/10.1080/09540261.2020.1713056

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